

```
let age = prompt("Enter your age:");

age = parseInt(age); // Convert input to integer

if (age >= 18) {
    console.log("You are eligible to vote.");
} else if (age > 0 && age < 18) {
    console.log("You are not eligible to vote yet.");
} else {
    console.log("Please enter a valid age.");
}
```

✓ 1. while Loop Example

javascript

Copy

Edit

```
let i = 1;

while (i <= 5) {
    console.log("While Loop:", i);
    i++;
}
```

✓ 2. for Loop Example

javascript

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```
for (let j = 1; j <= 5; j++) {
    console.log("For Loop:", j);
}
```



✓ 3. Function to Check Voting Eligibility

javascript

Copy

Edit

```
function isEligibleToVote(age) {  
  if (age >= 18) {  
    return "Eligible to vote.";  
  } else {  
    return "Not eligible to vote.";  
  }  
}  
  
console.log(isEligibleToVote(20)); // Output: Eligible to vote.
```

✓ 4. Function Without Parameters

javascript

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```
function sayWelcome() {  
  console.log("Welcome to JavaScript!");  
}  
  
sayWelcome(); // Output: Welcome to JavaScript!
```

javascript

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```
// Declare and initialize an array
let fruits = ["Apple", "Banana", "Mango"];

// Access elements
console.log("First fruit:", fruits[0]); // Apple

// Update an element
fruits[1] = "Orange";
console.log("Updated fruits:", fruits); // ["Apple", "Orange", "Mango"]

// Add a new element
fruits.push("Grapes");
console.log("After adding Grapes:", fruits); // ["Apple", "Orange", "Mango", "Grapes"]

// Loop through the array
console.log("List of fruits:");
for (let i = 0; i < fruits.length; i++) {
    console.log(fruits[i]);
}
```

```
// Initial array
let colors = ["Red", "Green", "Blue"];
console.log("Original array:", colors);

// push() - Add an item to the end
colors.push("Yellow");
console.log("After push (add to end):", colors); // ["Red", "Green", "Blue", "Yellow"]

// pop() - Remove the last item
let removedColor = colors.pop();
console.log("After pop (remove from end):", colors); // ["Red", "Green", "Blue"]
console.log("Popped item:", removedColor); // "Yellow"

// unshift() - Add an item to the beginning
colors.unshift("Pink");
console.log("After unshift (add to beginning):", colors); // ["Pink", "Red", "Green", "Blue"]

// shift() - Remove the first item
let shiftedColor = colors.shift();
console.log("After shift (remove from beginning):", colors); // ["Red", "Green", "Blue"]
console.log("Shifted item:", shiftedColor); // "Pink"
```

javascript

```
let weight = parseFloat(prompt("Enter your weight in kg:"));
let height = parseFloat(prompt("Enter your height in meters:"));

let bmi = weight / (height * height);
let roundedBMI = Math.round(bmi);

console.log("Your BMI is:", roundedBMI);
```

javascript

Copy

Edit

```
let weight = parseFloat(prompt("Enter your weight in kg:"));
let height = parseFloat(prompt("Enter your height in meters:"));

let bmi = weight / (height * height);
let roundedBMI = Math.round(bmi); // Rounds to the nearest whole number

console.log("Your BMI is:", roundedBMI);

// BMI Category Check
if (roundedBMI < 18.5) {
    console.log("You are underweight.");
} else if (roundedBMI >= 18.5 && roundedBMI < 25) {
    console.log("You have a normal weight.");
} else if (roundedBMI >= 25 && roundedBMI < 30) {
    console.log("You are overweight.");
} else {
    console.log("You are obese.");
}
```

javascript

Copy

Edit

```
let n = parseInt(prompt("Enter the number of terms:"));

let a = 0, b = 1, nextTerm;

console.log("Fibonacci Series:");

for (let i = 1; i <= n; i++) {
    console.log(a);
    nextTerm = a + b;
    a = b;
    b = nextTerm;
}
```

javascript

Copy

Edit

```
let weight = parseFloat(prompt("Enter your weight in kg:"));
let height = parseFloat(prompt("Enter your height in meters:"));

let bmi = weight / (height * height);
let roundedBMI = Math.round(bmi);

console.log("Your BMI is:", roundedBMI);
```

javascript

Copy

Edit

```
let str = prompt("Enter a word to check for palindrome:");
let isPalindrome = true;

for (let i = 0; i < str.length / 2; i++) {
    if (str[i] !== str[str.length - 1 - i]) {
        isPalindrome = false;
        break;
    }
}

if (isPalindrome) {
    console.log(str + " is a palindrome.");
} else {
    console.log(str + " is not a palindrome.");
}
```

javascript

Copy

Edit

```
let num = prompt("Enter a number:");
let originalNum = num; // Store the original number
let reversedNum = 0;
let remainder;

while (num > 0) {
    remainder = num % 10; // Get the last digit
    reversedNum = reversedNum * 10 + remainder; // Build the reversed number
    num = Math.floor(num / 10); // Remove the last digit from the number
}

if (originalNum == reversedNum) {
    console.log(originalNum + " is a palindrome number.");
} else {
    console.log(originalNum + " is not a palindrome number.");
}
```

DOM Manipulation

✓ Example 1: getElementById()

♦ HTML:

html

Copy

Edit

```
<!DOCTYPE html>
<html>
<body>
  <h1 id="myHeading">Hello!</h1>
  <button onclick="changeText()">Click Me</button>

  <script>
    function changeText() {
      var heading = document.getElementById("myHeading");
      heading.innerText = "You clicked the button!";
    }
  </script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>
  <h1 id="myTitle">This is a Title</h1>

  <script>
    const title = document.querySelector("#myTitle");
    title.textContent = "Title has been changed!";
  </script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>
  <p class="myPara">First paragraph</p>
  <p class="myPara">Second paragraph</p>
```



```
<script>
  const firstPara = document.querySelector(".myPara");
  firstPara.textContent = "First paragraph changed!";
</script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>
  <h2>This is a heading</h2>
  <p>This is a paragraph</p>

  <script>
    const firstHeading = document.querySelector("h2");
    firstHeading.textContent = "The heading has changed!";
  </script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>
  <div class="container">
    <p class="text">Paragraph 1</p>
    <p class="text">Paragraph 2</p>
  </div>

  <script>
    const firstText = document.querySelector(".container > .text");
    firstText.textContent = "First paragraph in container!";
  </script>
```

```
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<body>
  <p id="text">Watch me change color!</p>
  <button onclick="changeColor()">Change Color</button>

  <script>
    function changeColor() {
      document.getElementById("text").style.color = "red";
    }
  </script>
</body>
</html>
```

```
let box = document.getElementById("myBox");
box.style.backgroundColor = "yellow";
box.style.padding = "20px";
box.style.borderRadius = "10px";
```

```
<p id="message">Old message here</p>
<button onclick="updateMessage()">Update</button>

<script>
  function updateMessage() {
```

```
    document.getElementById("message").textContent = "New message updated!";
  }
</script>
```

```
<h2 id="heading">Welcome to JS</h2>
```

```
<script>
  let headingText = document.getElementById("heading").textContent;
  alert("Heading says: " + headingText);
</script>
```

```

<button onclick="changeImage()">Change Image</button>
```

```
<script>
  function changeImage() {
    document.getElementById("myImg").setAttribute("src", "image2.jpg");
  }
</script>
```

```
<a id="myLink" href="https://google.com">Go to Google</a>
<button onclick="changeLink()">Change Link</button>
```

```
<script>
  function changeLink() {
    document.getElementById("myLink").setAttribute("href", "https://openai.com");
  }
</script>
```

```
    document.getElementById("myLink").textContent = "Go to OpenAI";
  }
</script>
```

```
<button id="btn">Click me</button>
<p id="result"></p>
```

```
<script>
  document.getElementById("btn").addEventListener("click", function() {
    document.getElementById("result").textContent = "Button was clicked!";
  });
</script>
```

```
<p id="hoverText">Hover over me!</p>
```

```
<script>
  document.getElementById("hoverText").addEventListener("mouseover", function() {
    this.style.color = "green";
  });
</script>
```

```
<input type="text" id="inputBox" placeholder="Type something">
<p id="keyResult"></p>
```

```
<script>
  document.getElementById("inputBox").addEventListener("keydown", function(event) {
    document.getElementById("keyResult").textContent = "You pressed: " + event.key;
  });
</script>
```

```
</script>
```

```
<input type="text" id="textBox" placeholder="Press Enter">
<p id="messageBox"></p>
```

```
<script>
  document.getElementById("textBox").addEventListener("keydown", function(event) {
    if (event.key === "Enter") {
      document.getElementById("messageBox").textContent = "Enter key was pressed!";
    }
  });
</script>
```

jQuery

```
<!DOCTYPE html>
<html>
<head>
  <title>jQuery Example</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>
<body>
  <button id="myBtn">Click Me</button>

  <script>
    $(document).ready(function() {
      $("#myBtn").click(function() {
        alert("Hello! jQuery is working.");
      });
    });
  </script>
</body>
</html>
```

```
<p id="demo">Hello</p>
<button>Click</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#demo").text("Changed Text");
  });
});
</script>
```

```
<p class="info">Info 1</p>
<p class="info">Info 2</p>
<button>Update</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $(".info").text("Updated!");
  });
});
</script>
```

```
<p id="colorText">Change my color</p>
<button>Change Color</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#colorText").css("color", "blue");
  });
});
```

```
});  
</script>
```

```
<p id="box">I'm a box</p>  
<button>Style Me</button>
```

```
<script>  
$(document).ready(function() {  
  $("button").click(function() {  
    $("#box").css({  
      "color": "white",  
      "background-color": "black",  
      "padding": "10px"  
    });  
  });  
});  
</script>
```

```
<p id="txt">Original</p>  
<button>Change</button>
```

```
<script>  
$(document).ready(function() {  
  $("button").click(function() {  
    $("#txt").text("New Text");  
  });  
});  
</script>
```

```
<p id="htmlContent">Old <b>bold</b> text</p>  
<button>Replace</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#htmlContent").html("<i>Italic Text</i>");
  });
});
</script>
```

```
<div id="container">Welcome</div>
<button>Add Text</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#container").append("<p>New Paragraph</p>");
  });
});
</script>
```

```
<p id="rem">Remove me</p>
<button>Remove</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#rem").remove();
  });
});
</script>
```

```
<p id="animate">Watch me</p>
```



```
<button>Hide</button>
<button>Show</button>
```

```
<script>
$(document).ready(function() {
  $("button").eq(0).click(function() {
    $("#animate").hide(1000);
  });
  $("button").eq(1).click(function() {
    $("#animate").show(1000);
  });
});
</script>
```

```
<div id="panel" style="display:none; background:lightblue;">This is a panel</div>
<button>Toggle Panel</button>
```

```
<script>
$(document).ready(function() {
  $("button").click(function() {
    $("#panel").slideToggle();
  });
});
</script>
```

html

Copy

Edit

```
<!DOCTYPE html>
<html>
<head>
  <title>Final Demo</title>
  <!-- Link to external CSS -->
  <link rel="stylesheet" href="style.css">
  <!-- Link to jQuery Library -->
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <!-- Link to external JS -->
  <script src="script.js" defer></script>
</head>
<body>

<div id="box">Click Me</div>

</body>
</html>
```

css

Copy

Edit

```
#box {  
  width: 200px;  
  height: 200px;  
  background: lightcoral;  
  text-align: center;  
  line-height: 200px;  
  margin: 20px auto;  
}
```

Save this file in the same folder as your HTML file.

✓ Step 3: External JS File (script.js)

javascript

Copy

Edit

```
$(document).ready(function() {  
  $("#box").click(function() {  
    $(this).css("background", "lightgreen").text("Changed!");  
  });  
});
```



html

Copy

Edit

```
<!DOCTYPE html>
<html>
<head>
  <title>Final Demo</title>
  <!-- Link to external CSS -->
  <link rel="stylesheet" href="style.css">
  <!-- Link to external JS -->
  <script src="script.js" defer></script>
</head>
<body>

<div id="box">Click Me</div>

</body>
</html>
```

css

Copy

Edit

```
#box {
  width: 200px;
  height: 200px;
  background: lightcoral;
  text-align: center;
  line-height: 200px;
  margin: 20px auto;
}
```

javascript

Copy

Edit

```
document.addEventListener("DOMContentLoaded", function() {
  // Select the box element
  var box = document.getElementById("box");

  // Add a click event listener
  box.addEventListener("click", function() {
    // Change the background color and text on click
    box.style.background = "lightgreen";
    box.textContent = "Changed!";
  });
});
```

html

Copy

Edit

```
<!DOCTYPE html>
<html>
<head>
  <title>Final Demo</title>
  <!-- Link to external CSS -->
  <link rel="stylesheet" href="style.css">
  <!-- Link to external JS -->
  <script src="script.js" defer></script>
</head>
<body>

<button id="box">Click Me</button>

</body>
</html>
```